Human Activity Recognition Using Accelerometer Signals

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Abstract. This assignment addresses the development of a human activity recognition (HAR) system using accelerometer signals collected from mobile phones. The project includes signal preprocessing, feature extraction with sliding windows, and training of classification models under a Leave-One-Subject-Out (LOSO) cross-validation setup.

Keywords: Activity Recognition · Accelerometer · Machine Learning · Feature Extraction · LOSO

1 Introduction

Human Activity Recognition (HAR) leverages data from wearable or mobile sensors to predict physical activities. With the rise of mobile technologies, HAR systems are increasingly deployed for healthcare, fitness, and smart environments. Accurate HAR on unconstrained, real-world data persists as a challenge due to sensor noise, variable placement, and the subtlety of user movements. This work, involves preprocessing raw sensor data from an accelerometer, extracting statistical and spectral features from time windows, and training classification models on seven everyday activities including SVM, Random Forests and MLPs using a Leave-One-Subject-Out strategy for generalization evaluation.

2 Research Overview

Our dataset, derived from Shoaib et al. [3], contains accelerometer readings from 10 participants performing seven physical activities (biking, downstairs, jogging, sitting, standing, upstairs, walking), with smartphones placed at various body positions.

Participants carried smartphones in various positions (Left/Right pocket and wrist) during each activity, with the sensor orientation fixed (portrait) for the pockets, and sampling frequency set at 50 Hz. Each entry contains timestamps, three-axis acceleration, and a categorical activity label. We standardized the analysis to the "left pocket" location. In total, each participant contributed approximately 21 minutes of continuous labeled data, yielding 63000 samples per participant for all activities.

Our research questions include:

- Can we reliably classify physical activity using only accelerometer data from a specific smartphone position?
- Which preprocessing and feature extraction steps improve classification accuracy?
- How do different models perform under the LOSO validation scheme?
- Can we improve classification by grouping similar activities?
- How does the sensor placement affect classification performance?

3 System Design

3.1 Pipeline Overview

The pipeline involves preprocessing raw sensor data, cleaning, extracting statistical and spectral features from time sliding - windows, and training classification models including SVM, Random Forests and MLPs using a Leave-One-Subject-Out strategy for generalization evaluation. Numerous grid searches were performed to optimize hyperparameters for each model. At the end, we evaluate the models' performance using confusion matrices and classification accuracy metrics. We focus as well on combining similar activities and the evaluation of different sensor placement when the models are trained on a specific position and tested on another.[1]

Figure 1 depicts the architecture.

- **Preprocessing:** Clean data, synchronize, compute vector magnitude, and handle missing or corrupted values.
- **Segmentation:** Divide the signal into overlapping sliding windows.
- **Feature Extraction:** Compute both time and frequency domain features per window.
- **Classification:** Fit models with hyperparameter tuning; test on unseen subjects using LOSO.

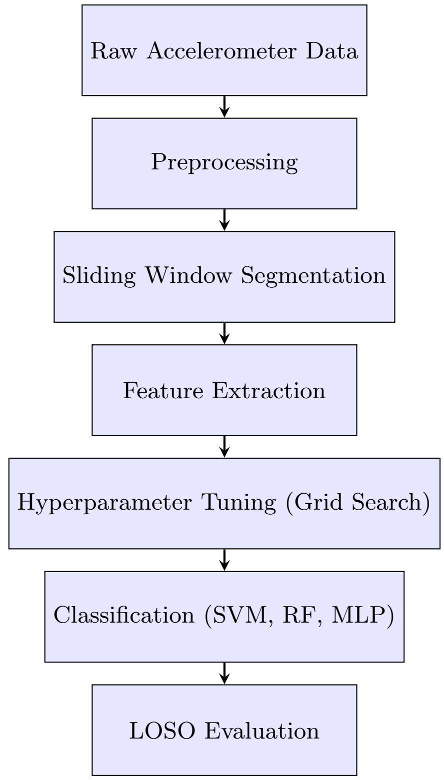


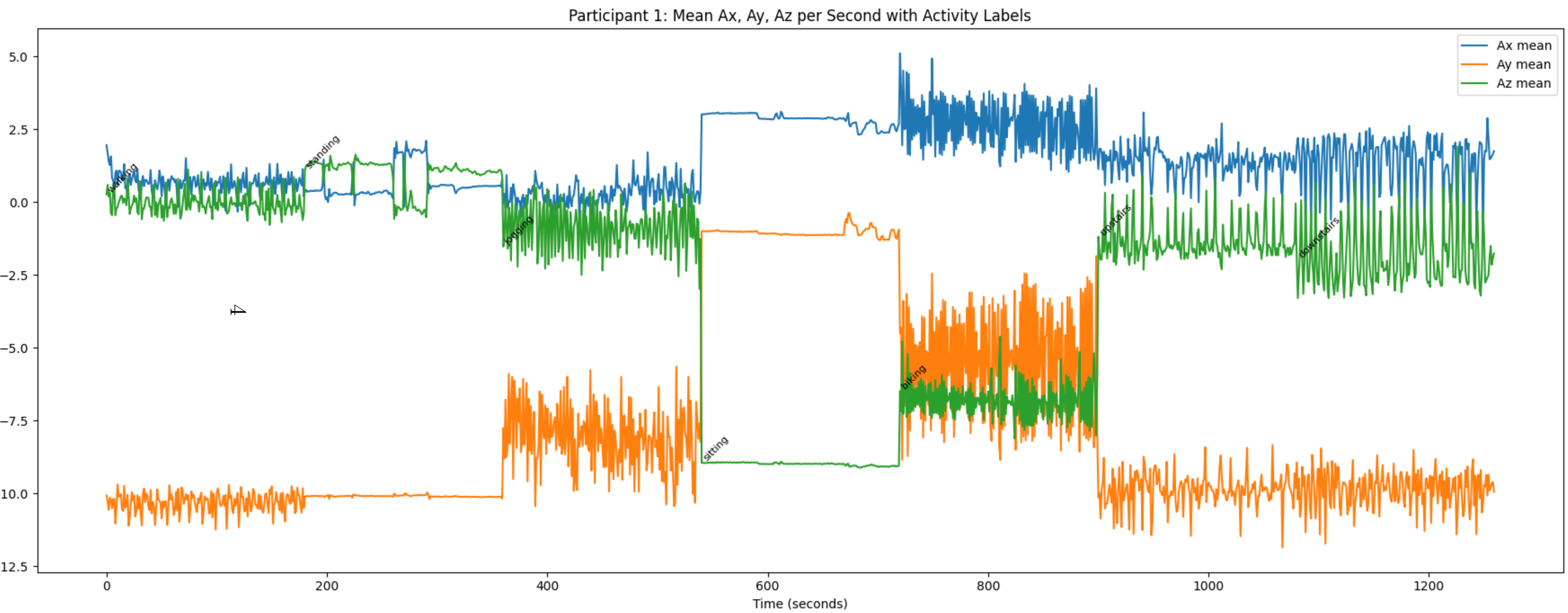
Fig. 1. Outlines the high-level architecture of our system pipeline. It begins with preprocessing raw accelerometer signals, followed by sliding window segmentation and feature extraction. Finally, multiple classification models are trained and evaluated using LOSO cross-validation.

4 Data Preprocessing

4.1 Raw Data Visualization

Initial visualizations were performed to understand signal distribution and anomalies. Accelerometer data from the left pocket position was selected. Signals were plotted over time to confirm activity transitions and to inspect potential sensor errors. Visual inspection is indispensable for identifying sensor issues, label mismatch, or outliers.

1. Plotted raw values (Ax, Ay, Az) across time, color-coded by activity Fig.2.
2. Examined per-axis histograms for each activity Fig.3, observing clear signal patterns (e.g., high Az for sitting/standing, more dynamic Ax for jogging).
3. Quantified duration and balance of classes, confirming uniform label frequencies.


Participant 1: Mean Ax, Ay, Az per Second with Activity Labels
Ax mean
Ay mean
Az mean
walking
standing
jogging
sitting
biking
upstairs
downstairs
Time (seconds)
0
200
400
600
800
1000
1200
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
−12.5

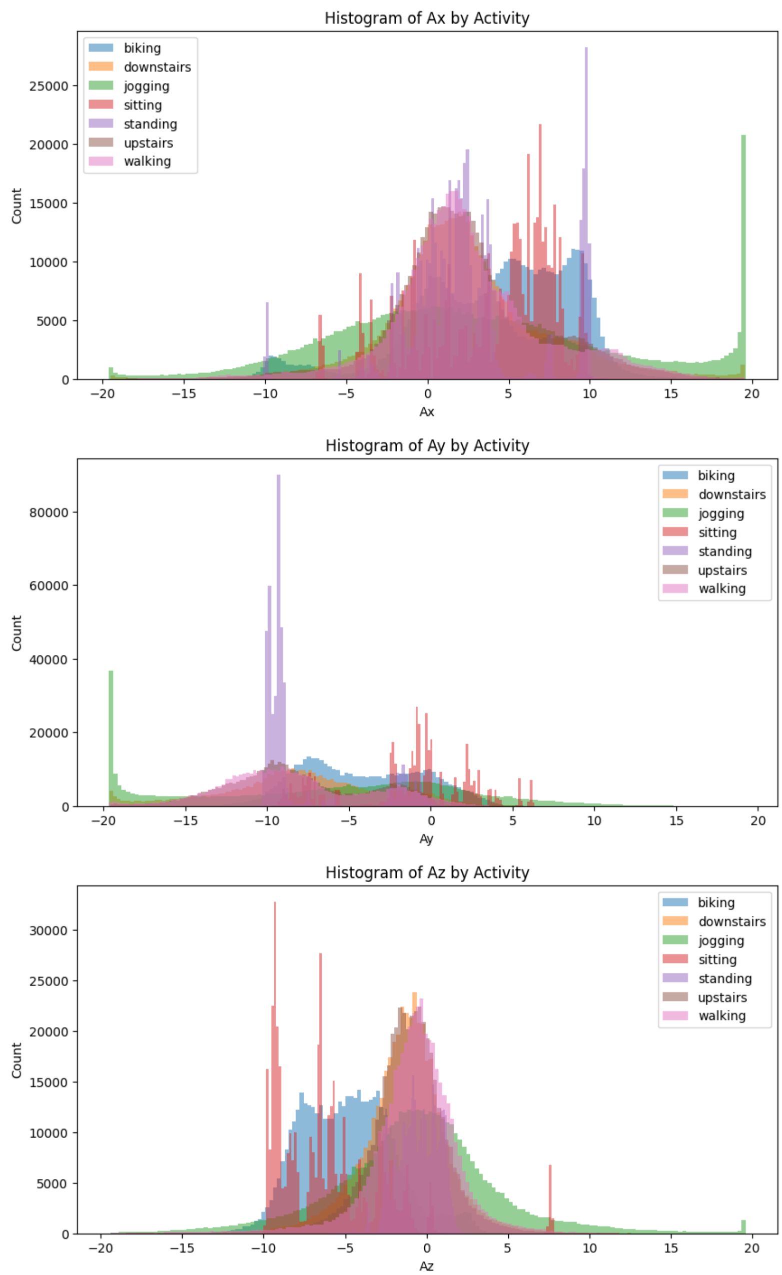


Fig. 3. Histogram showing the distribution of Ax

4.2 Data Preprocessing and Quality Assurance

Column Selection and Concatenation: We retained only time, Ax, Ay, Az, position of the sensor, participant ID, and activity label. The dataset included other variables from gyroscope and magnetometer, but we focused solely on the accelerometer data for this assignment.

4.3 Exploratory Data Analysis & Visualization of Accelerometer Signals

To validate data quality and inspect class separability, we:

– Plotted histograms for the distribution of Ax, Ay, Az per activity. Biking and jogging, for instance, show broader spreads, reflecting higher dynamics.
– Calculated mean, median, and variance for each activity and axis, confirming domain-expectant values, e.g., Az $\approx$ 9.8 m/s^2 in stationary behaviors (gravity).
– Plotted the per-second rolling mean of the magnitude signal, annotated with true activity, to visualize transitions on a participant.

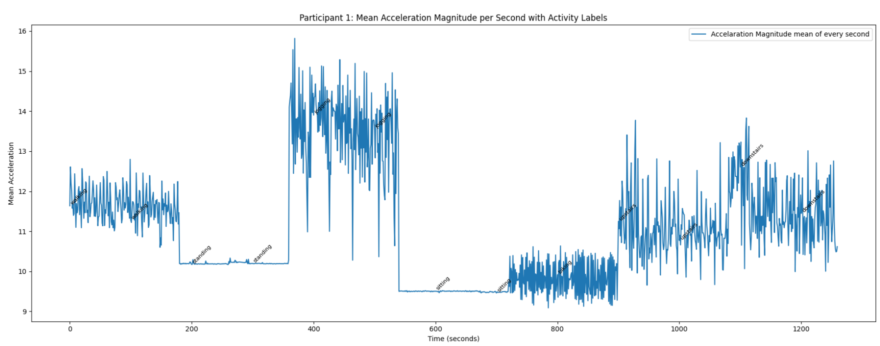


Fig. 4. Rolling mean of magnitude over time for participant 1, with true activity labels.

Sensor Magnitude: The 3D acceleration was reduced to a single-magnitude time-series as:

$$a(t) = \sqrt{Ax(t)^2 + Ay(t)^2 + Az(t)^2}$$

This step reduces dimensionality and can increase robustness to orientation, though at the cost of some information.

Error Handling and Outlier Removal: Sensor glitches can produce extreme outlier values (often $>$ 1000 m/s^2). These are replaced by the most recent non-anomalous value to prevent distortion of downstream statistics. No missing values were found after this step.

The preprocessing involved in short the following steps:

1. **Position Selection**: Only data from the "left pocket" was retained for consistency.
2. **Magnitude Conversion**: The 3D accelerometer signal was converted to 1D magnitude.
3. **Sensor Error Correction**: All magnitude values exceeding 1000 (indicative of sensor errors) were replaced with the previous valid value.

5 Feature Extraction

To transform raw sensor data into meaningful inputs for machine learning models, we applied a structured feature extraction pipeline. This process began with segmenting the magnitude of the accelerometer signal using a sliding window approach, followed by the computation of statistical and spectral features that capture both the time-domain and frequency-domain characteristics of human activities.

5.1 Sliding Window Segmentation

The sliding window approach allows us to capture temporal dynamics and local patterns in the accelerometer data. Each window overlaps with the previous one, ensuring continuity and capturing transitions between activities. The overlapping property was chosen to ensure that transitions between activities are captured effectively, allowing the model to learn from both stationary and dynamic segments. The windowing process was implemented as follows:

- The window size was set to 20 seconds, with a stride of 1 second.
- Each window contains 1000 samples (20 seconds at 50 Hz).
- The label for each window was determined by the majority activity within that window.

5.2 Statistical Measures as Features

- The following features were extracted:
  - Mean
  - Standard deviation
  - Skewness
  - Maximum
  - Minimum
  - Range (max - min)
  - Power spectral density (Welch's method)
- Features were standardized using z-score normalization.

5.3 Selecting Power Spectral Density Features

In machine learning applications involving accelerometer data, the use of Power Spectral Density (PSD) as a feature extraction method is highly valuable for capturing the underlying frequency characteristics of human activity signals. PSD quantifies how the power of a signal is distributed across different frequency components, which is particularly useful for recognizing and distinguishing between activities with different motion patterns and intensities. Among the PSD-derived features, the spectral centroid and spectral bandwidth (or spread) are especially informative. The spectral centroid, defined as the weighted average of the frequency components present in the signal, indicates the "center of mass" of the frequency spectrum and is often associated with the perceived frequency or overall tempo of motion. The spectral bandwidth, calculated as the standard deviation around this centroid, reflects the concentration or dispersion of frequency content, capturing the complexity or variability of movement. These features help differentiate activities such as walking, running, or stationary states, as they exhibit distinct frequency distributions. Selecting the maximum power, bandwidth, and centroid as features provides a compact yet powerful representation of the frequency domain, enabling more accurate classification of activity types. Furthermore, these features are computationally efficient and interpretable, making them ideal for real-time applications in wearable systems or mobile health monitoring. The removal of the 0Hz component is crucial, as it represents the DC offset of the signal, which does not contribute to distinguishing between different activities. By focusing on the non-zero frequency components, we enhance the model's ability to capture relevant motion dynamics without being influenced by static offsets.

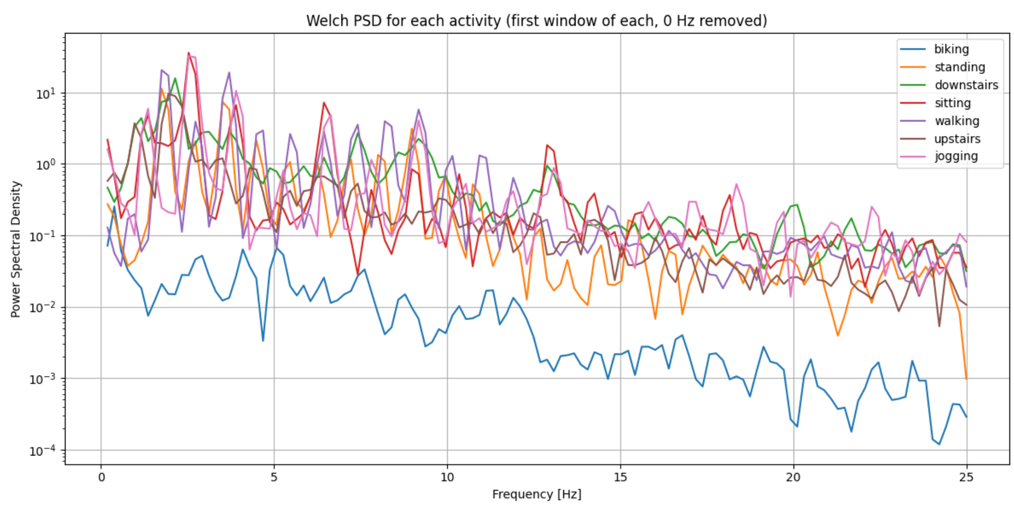


Fig. 5. Welch's Power Spectrum from participant 1, color-coded by activity.

6 Model Construction

We trained and evaluated three supervised learning models for activity recognition using features derived from the time and frequency domains of accelerometer data. The models include a Support Vector Machine (SVM) with a radial basis function (RBF) kernel or polynomial kernel with various degrees, a Random Forest classifier, and a Multilayer Perceptron (MLP). All models were trained using a Leave-One-Subject-Out (LOSO) cross-validation approach, in which each fold involved training on data from $n-1$ subjects and testing on the remaining subject. This approach ensures the generalizability of the models across individuals. Hyperparameters for each model were optimized using grid search strategies, and model performance was evaluated using confusion matrices and overall classification accuracy. We trained the following models:

6.1 Support Vector Machine (SVM) Classifier

Support Vector Machines are powerful margin-based classifiers that perform well in high-dimensional spaces. The use of the RBF kernel allows the model to capture non-linear decision boundaries, making it suitable for complex patterns in the feature space, such as those arising from human movement data. The RBF kernel transforms the feature space by mapping inputs into a higher-dimensional space where a linear separation is feasible.

Model Training and Evaluation For each LOSO fold, the SVM model was trained on the 8 selected features, including time-domain statistics and frequency-domain metrics derived from the Power Spectral Density (PSD). The performance was recorded in terms of precision, recall, and F1-score for each activity class. The model generally performed well in distinguishing between dynamic activities (e.g., walking, running), where frequency features were more discriminative.

Grid Search & Fine Tuning Hyperparameter tuning was performed using grid search over a predefined range for the regularization parameter C ,the RBF kernel parameter γ and the polynomial kernel. The search space included $C \in [0.1, 1, 10, 10]$ and $\gamma \in [0.001, 0.01, 0.1]$. We used the polynomial kernel as well with degrees of 2 and 3. The best combination of parameters was selected based on cross-validation performance on the training folds in each LOSO iteration. Unlikeky the RBF kernel, the polynomial kernel allows for a more interpretable decision boundary, which can be beneficial for understanding how different features contribute to classification decisions, but it can be really computationally expensive for grid searching and using the LOSO training.

Confusion Matrices Using the optimized SVM configuration, the model achieved a mean test accuracy of 0.810 across the LOSO folds. The confusion matrices showed clear separation between dynamic activities such as walking and running, with minor misclassifications observed in more subtle transitions like standing vs. sitting. The use of frequency-domain features such as spectral centroid and bandwidth likely contributed to the model's ability to distinguish between high- and low-intensity movements.

Table 1. SVM Classification Report (Best Parameters: `C=0.1`, `gamma=0.01`, `kernel='poly'`) with 3-fold cross-validation

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.74	0.98	0.84	905
Downstairs	0.80	0.79	0.79	895
Jogging	1.00	0.96	0.98	905
Sitting	0.48	0.44	0.46	895
Standing	0.48	0.50	0.49	900
Upstairs	0.78	0.70	0.74	895
Walking	0.74	0.62	0.67	886
Accuracy		0.71		
Macro Avg	0.72	0.71	0.71	6281
Weighted Avg	0.72	0.71	0.71	6281

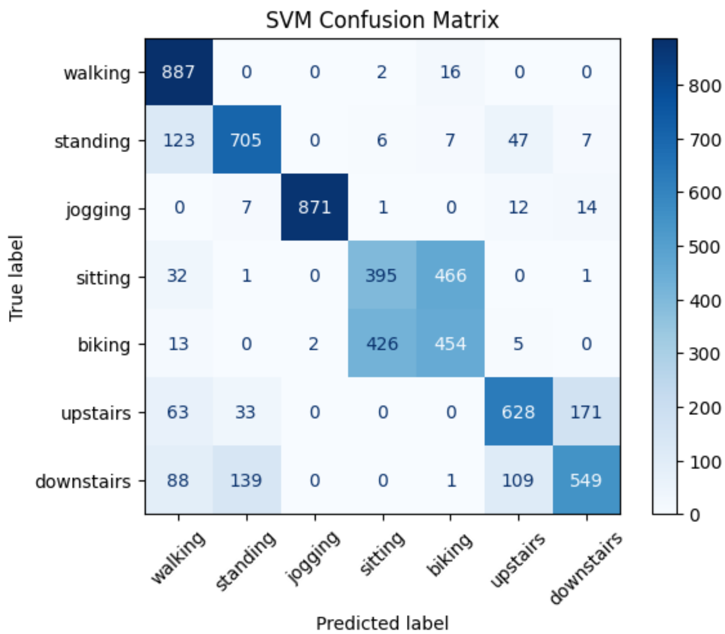


Fig. 6. Confusion matrix for the SVM classifier, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

6.2 Random Forest Classifier

Random Forests are ensemble-based classifiers that aggregate predictions from multiple decision trees to improve accuracy and robustness. They are well-suited for handling mixed-type feature sets and provide insights into feature importance, which is useful for understanding the discriminative power of time vs. frequency-domain features.

Model Training and Evaluation The Random Forest model was trained using 100 decision trees as a baseline. The model performed competitively across all LOSO folds, with particularly strong performance on static activities due to the ensemble's ability to average out noise and variance in sensor signals.

Grid Search & Fine Tuning For the Random Forest classifier, hyperparameter optimization was performed using grid search to identify the best model configuration. The parameter grid explored the following values:

- `n_estimators`: {100,150 ,200}
- `max_depth`: {None, 10, 15, 20}

– `min_samples_split`: {5, 10,15,20}
– `max_features`: {`sqrt`,`log2`}
– `criterion`: {`entropy`}

The search was conducted using cross-validation within the training folds of each LOSO iteration. The best results were typically obtained with `n_estimators = 200`, `max_depth = 10`, and `min_samples_split = 5`, using the `entropy` criterion for information gain. This configuration offered a balance between model complexity and generalization, and yielded consistent performance across subjects. Additionally, Random Forests provided useful insights into feature importance [4], supporting model interpretability.

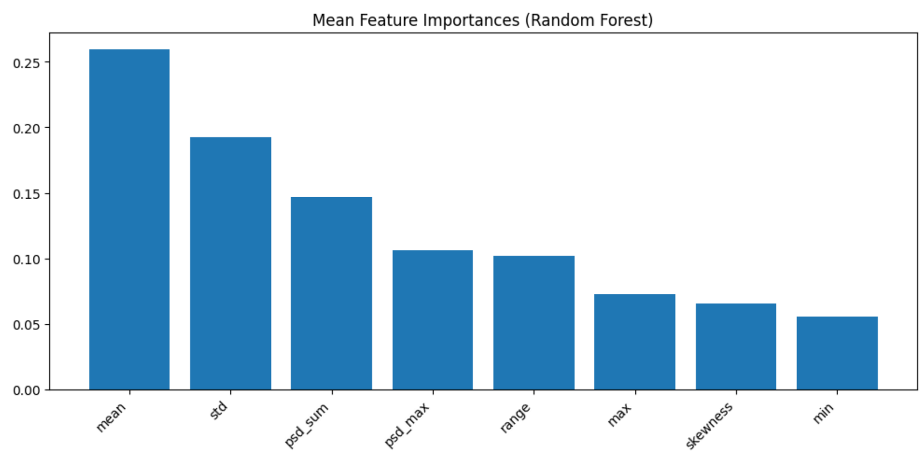


Fig. 7. Feature importance scores from the Random Forest model, indicating the relative contribution of each feature to the classification task.

What we immediately notice is that the most important features are the mean and standard deviation of the magnitude signal, followed by the spectral sum, spectral maximum and range. This suggests that both time-domain statistics and frequency-domain characteristics are crucial for distinguishing between activities.

Confusion Matrices The confusion matrices revealed that the Random Forest achieved high accuracy on both dynamic and static activities. However, minor confusion occurred between similar activities (e.g., walking upstairs vs. walking downstairs vs. walking), likely due to overlapping frequency distributions.

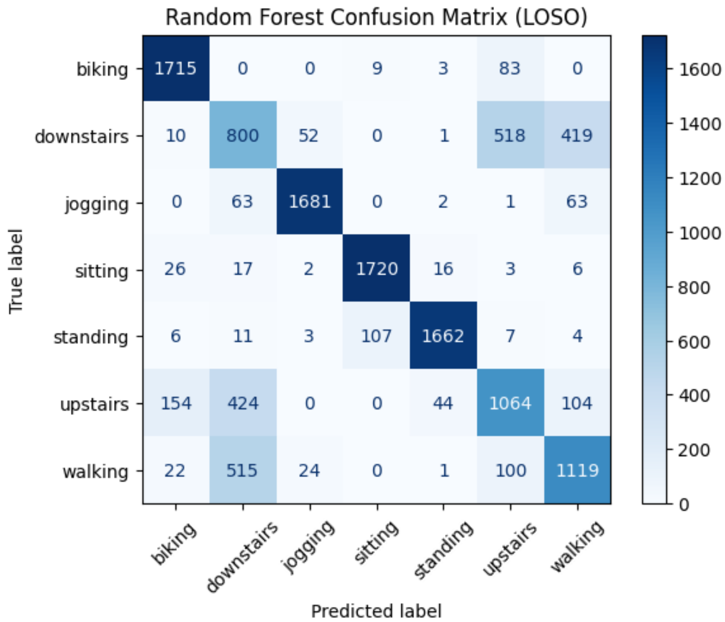


Fig. 8. Confusion matrix for the Random Forest classifier, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

6.3 Multilayer Perceptron (MLP) Classifier

The Multilayer Perceptron (MLP) is a feedforward neural network architecture capable of learning complex, non-linear patterns in data. It consists of fully connected layers of neurons with non-linear activation functions, trained using backpropagation. MLPs are particularly effective for classification tasks involving multi-dimensional feature sets such as those derived from accelerometer signals.

Model Training and Evaluation The MLP model was implemented with one to two hidden layers, each containing 10 to 50 neurons. The ReLU (Rectified Linear Unit) activation function was applied to all hidden layers to introduce non-linearity, while the output layer used softmax for multi-class classification. The model was trained using stochastic gradient descent (SGD) with momentum, and dropout regularization was applied to mitigate overfitting [2]. Training was conducted over multiple LOSO folds, with early stopping used to prevent overfitting on the training data. The model demonstrated good generalization when trained with appropriately tuned hyperparameters and sufficient data per fold.

Grid Search & Fine Tuning Hyperparameter tuning for the MLP involved grid search over the following parameters:

- `hidden_layer_sizes`: {(20,), (10,10), (20, 20),(50,50)}
- `learning_rate_init`: {0.001, 0.005, 0.01}
- `activation`: {`relu`, `tanh`}
- `solver`: {`adam`, `sgd`}
- `alpha`: {0.0001, 0.001, 0.01}
- `momentum`: { 0.7, 0.9, 0.95}
- `max_iteration`: {500, 1000,10000}

One of the good configurations was `hidden_layer_sizes = (50, 50)`, `solver = adam`, `max_iteration`: {10000} and `momentum = 0.9`, which provided stable and accurate convergence across most subjects. These parameters enabled the model to learn effectively without significant overfitting, achieving an accuracy of 0.77 but perfoming really bad in some activities such as downstairs and walking. In our case, our goal was to search for a good confisuratioin using the solver `sgd` and searching for the best `momentum` parameter. So, we concluded that the best configuration for the MLP was `hidden_layer_sizes = (20, 10)`, `solver = sgd`, `max_iteration`: {500} and `momentum = 0.95`. This achieved a mean accuracy of 67 % training with LOSO folds and test accuracy of 0.71. It performed really differently on different subjects, with some achieving up to 75% accuracy while others were below 60%.

Table 2. Classification report for Participant 1 (Accuracy: 0.750)

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.85	0.59	0.69	905
Downstairs	0.64	0.39	0.49	905
Jogging	0.93	0.95	0.94	905
Sitting	0.56	0.70	0.62	895
Standing	0.60	0.39	0.47	900
Upstairs	0.59	0.52	0.55	895
Walking	0.41	0.78	0.53	895
Accuracy			0.62	6300
Macro Avg	0.65	0.62	0.61	6300
Weighted Avg	0.65	0.62	0.61	6300

Table 3. Classification report for Participant 10 (Accuracy: 0.616)

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.84	0.90	0.87	905
Downstairs	0.62	0.79	0.70	905
Jogging	0.94	0.95	0.94	905
Sitting	0.57	0.93	0.71	895
Standing	0.79	0.27	0.40	900
Upstairs	0.72	0.42	0.53	895
Walking	0.53	0.59	0.56	895
Accuracy			0.69	6300
Macro Avg	0.72	0.69	0.67	6300
Weighted Avg	0.72	0.69	0.67	6300

Table 4. MLP Classification Report on Test Set

Activity	**Precision**	**Recall**	**F1-Score**	**Support**
Biking	0.95	0.59	0.73	181
Downstairs	0.64	0.60	0.62	171
Jogging	1.00	0.97	0.99	181
Sitting	0.83	0.93	0.88	179
Standing	0.96	0.99	0.98	180
Upstairs	0.66	1.00	0.80	179
Walking	0.96	0.76	0.85	170
Accuracy		0.838		
Macro Avg	0.86	0.84	0.83	1241
Weighted Avg	0.86	0.84	0.84	1241

Confusion Matrices Confusion matrices generated from the LOSO folds indicated that the MLP performed well on dynamic activities (Jogging, Biking), and the misclassifications occurred between similar classes such as walking and walking upstairs or downstair. Also, the static activities (Sitting, Standing) were often confused due to their similar acceleration patterns. Despite its flexibility, the MLP was more sensitive to training data volume and initialization compared to the SVM and Random Forest classifiers. Corresponding confusion matrices were computed for each model using the LOSO validation strategy.

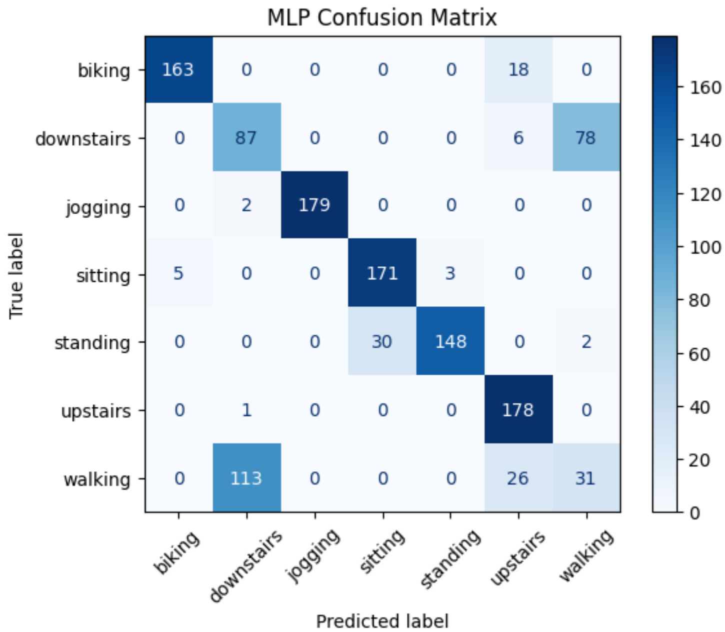


Fig. 9. Confusion matrix for the MLP classifier using the ADAM solver, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

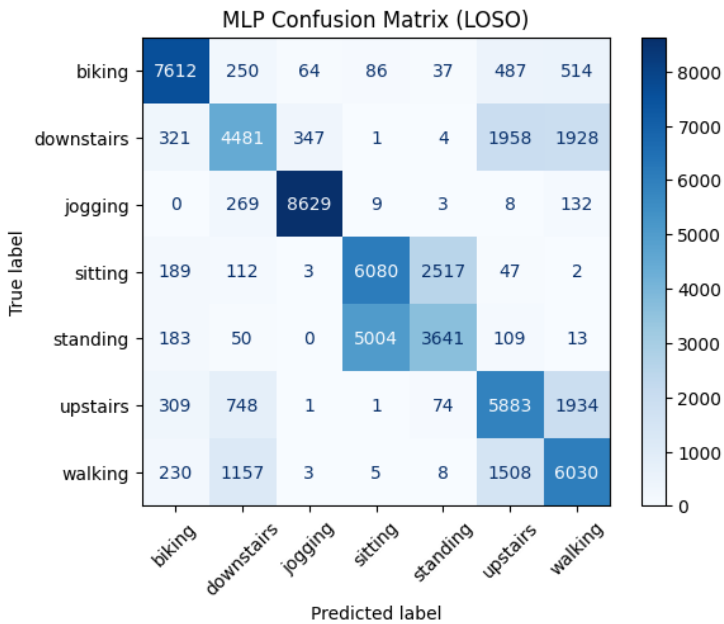


Fig. 10. Confusion matrix for the MLP classifier using the SGD solver, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

7 Selecting the Best Classifier

To determine the most suitable classifier for our application, we compared the performance of all three models based on averaged LOSO results. Overall, the Support Vector Machine achieved the highest classification accuracy, particularly in distinguishing high-frequency dynamic activities, thanks to its ability to leverage the discriminative power of PSD-based features. The Random Forest offered robust and interpretable results with minimal tuning, making it a strong candidate for deployment in resource-constrained settings. The MLP demonstrated potential in modeling complex feature interactions but required more extensive training and was more sensitive to hyperparameters and data distribution. Based on this analysis, the SVM is recommended for scenarios prioritizing accuracy, while the Random Forest may be preferred for simplicity and interpretability. Future work could include ensembling these models or using feature selection techniques to further enhance performance.

7.1 Confusion Matrix and Classification Accuracy

- **SVM**: Achieved an average accuracy of **72.9%**.
- **Random Forest**: Achieved an accuracy of **77.5%**.
- **MLP**: Achieved an accuracy of **83.8%**.

7.2 Analysis of Misclassifications

These confusions likely arise from:

- Similar signal patterns in transitional or stationary activities.
- The positioning of the sensor (pocket-based) causing overlapping acceleration features.
- Transforming the 3-dimension data to 1 dimension magnitude (loss of information, gain on computing time).

Particular confusions:

- **Walking, Upstairs, Downstairs:** All involve vertical/horizontal oscillation, making frequency and amplitude similar; 3D magnitude transformation loses body orientation.
- **Sitting vs. Standing:** Both are quasi-static, gravity-dominated activities.

Feature selection attempts to address this; further gains may require leveraging gyroscope, additional sensor data, or personal calibration.

7.3 Grouped-Activity Analysis and Updated Improvements

When we group similar activities (e.g., walking + stairs up + stairs down), and combine sitting + standing labels and training again the Random Forest Classifier, accuracy increases (**to about** 77.5%)—demonstrating that label granularity plays a significant role in achievable accuracy.

Based on confusion analysis, we grouped the following:

- **Group A:** Upstairs, Downstairs
- **Group B:** Sitting, Standing

Table 5. Classification Report with Grouped Activities

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.87	0.87	0.87	905
DownUps	0.78	0.84	0.81	1790
Jogging	1.00	0.99	0.99	905
SitStand	0.99	0.96	0.98	1795
Walking	0.67	0.61	0.64	886
Accuracy			0.87	6281
Macro Avg	0.86	0.85	0.86	6281
Weighted Avg	0.87	0.87	0.87	6281

Random Forest managed to achieve this performance based on the grouped activities and showed a significant improvement in classification accuracy.

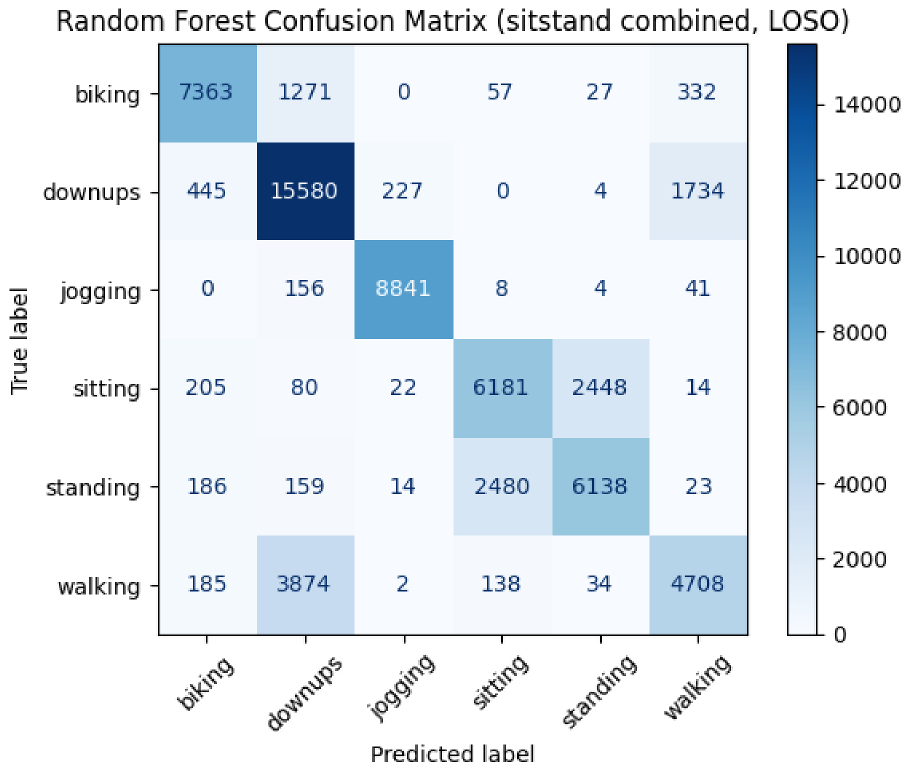


Fig. 11. Confusion matrix for the Random Forest classifier with grouped activities, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

Retraining the optimal SVM (polynomial kernel degree of 2 and gamma equal to 0.125) using these grouped labels, the updated classification accuracy increased to **87.8%**, suggesting that activity similarity can be exploited to simplify the recognition task with higher robustness. This indication of improved performance suggests that the model can generalize better when activities are semantically similar, reducing the complexity of the classification task. In our case, upstairs and downstairs activities share similar acceleration patterns, as do sitting and standing. So, it was sensible to group them together without losing too much information and gaining on performance.

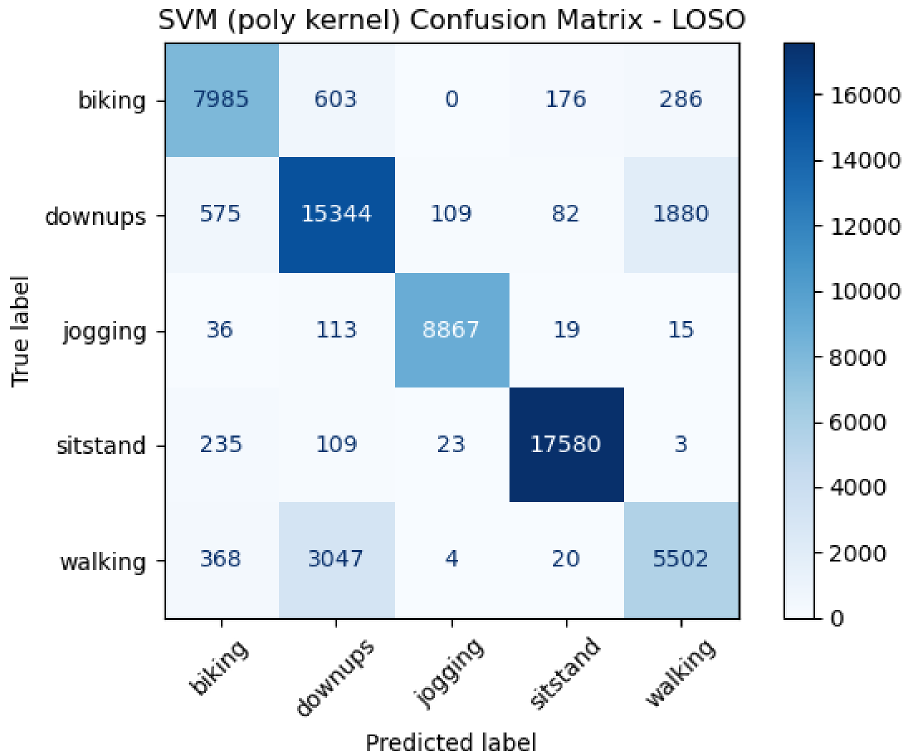


Fig. 12. Confusion matrix for the Random Forest classifier with grouped activities, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

Table 6. Classification Report (Optimal SVM Classifier on grouped Activities)

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.87	0.88	0.88	9050
Downups	0.80	0.85	0.82	17990
Jogging	0.98	0.98	0.98	9050
Sitstand	0.98	0.98	0.98	17950
Walking	0.72	0.62	0.66	8941
Accuracy		0.88		
Macro Avg	0.87	0.86	0.87	62981
Weighted Avg	0.88	0.88	0.88	62981

Moreover, the multiple layer perceptron (MLP) classifier, when retrained with the grouped activities, achieved a mean accuracy of **72.3 %** across the LOSO folds, indicating that it can also benefit from this grouping strategy, although it still lags behind the SVM and Random Forest in terms of overall performance.

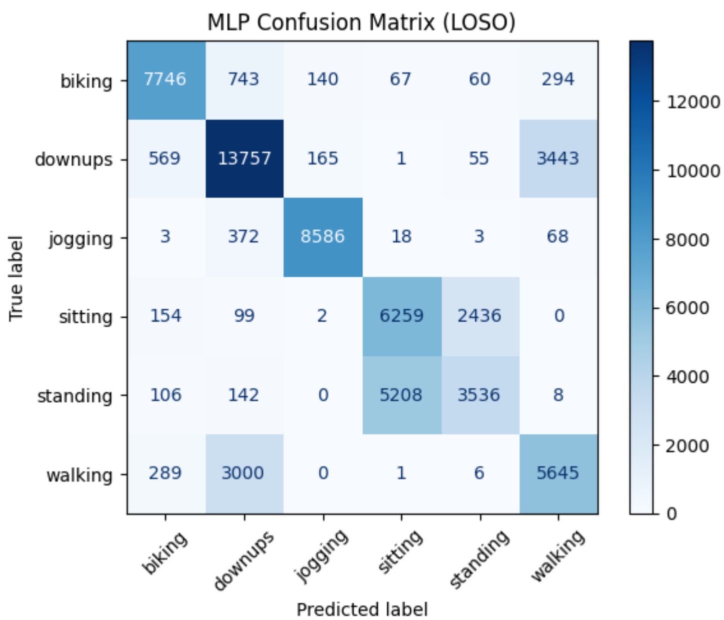


Fig. 13. Confusion matrix for the MLP classifier with grouped activities, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

Lastly, we show the results of the Leave-One-Subject-Out (LOSO) cross-validation for each classifier, which provides a more granular view of how each model performs across different subjects. This is crucial for understanding the generalization capabilities of the models in real-world scenarios.

7.4 Sensor Placement Evaluation Using the Best Classifier

Using the best-performing model (Random Forest), trained on all 10 subjects: We tested models trained on left pocket and evaluated on right pocket. Accuracy was sustained at **78.2%** for the right pocket, indicating that the model generalizes well across similar pocket placements. The misclassifications were mainly between the static activities (sitting vs. standing). However, when testing on

Table 7. LOSO Accuracy Comparison Across Classifiers

Participant	Random Forest	MLP	SVM (poly)
1	0.906	0.788	0.888
2	0.833	0.701	0.886
3	0.802	0.699	0.870
4	0.848	0.700	0.850
5	0.843	0.708	0.848
6	0.884	0.723	0.910
7	0.900	0.730	0.914
8	0.842	0.781	0.882
9	0.867	0.750	0.866
10	0.808	0.649	0.863
Mean Accuracy	**0.853**	**0.723**	**0.878**

wrist locations, the classification accuracy might drop due to heterogeneous signal characteristics and orientation differences. A better estimation of the differences might be prediction intervals and statistical tests, but exceeds the scope of this assignment.

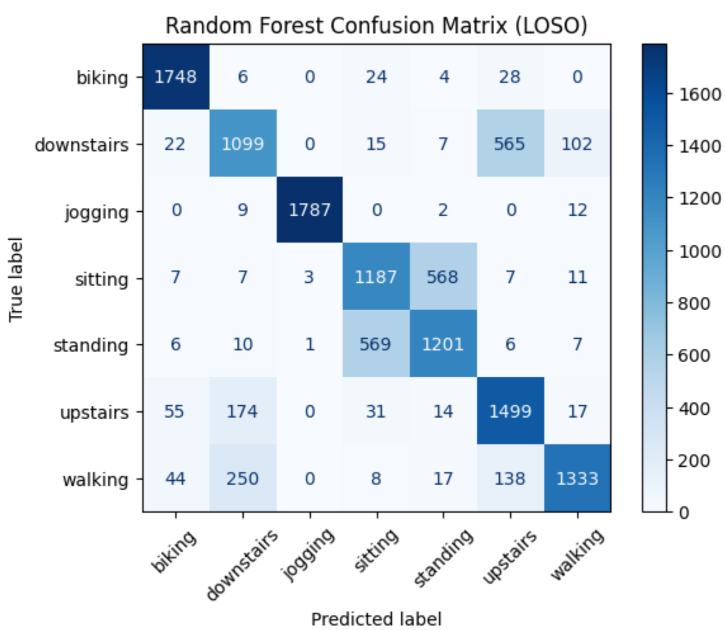


Fig. 14. Confusion matrix for the Random Forest classifier trained on left pocket data and tested on right pocket data, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

This highlights the critical importance of fixed or known device positions in wearable HAR systems and the need for multiple position data to train the optimal models.

Table 8. Classification Report for Sensor Placement Evaluation (Random Forest)

Activity	**Precision**	**Recall**	**F1-score**	**Support**
Biking	0.82	0.84	0.83	181
Downstairs	0.94	0.66	0.78	181
Jogging	1.00	0.99	0.99	181
Sitting	0.08	0.01	0.02	179
Standing	0.51	0.99	0.67	180
Upstairs	0.83	0.89	0.86	179
Walking	0.86	0.95	0.90	179
Accuracy			0.76	1260
Macro Avg	0.72	0.76	0.72	1260
Weighted Avg	0.72	0.76	0.72	1260

This performance degradation reflects the sensitivity of accelerometer-based models to sensor orientation and body location, emphasizing the importance of consistent device positioning in Human Activity Recognition (HAR) systems and the similarity of positioning the smartphone in the left or right pocket. The enormous difference in performance when testing different pockets were the activies of sitting and standing which were totally misclassified when checking the other perfomance metrics such as precision, recall and F1-score.

The optimal SVM classifier, when trained on left pocket data and tested on right pocket data, achieved an accuracy of **72.8%**, with similar misclassifications as the Random Forest model and worse results on seperating the sitting and standing.

Table 9. LOSO Accuracy per Participant (SVM, Left→Right Transfer)

Participant	Accuracy
1	0.709
2	0.815
3	0.769
4	0.711
5	0.727
6	0.835
7	0.545
8	0.727
9	0.812
10	0.625
Mean Accuracy	**0.728**

Table 10. Classification Report – SVM (Left→Right Transfer Test Set)

Activity	Precision	Recall	F1-score	Support
Biking	0.82	0.98	0.89	181
Downstairs	0.93	0.90	0.91	181
Jogging	0.99	1.00	1.00	181
Sitting	1.00	0.01	0.01	179
Standing	0.50	0.99	0.66	180
Upstairs	0.96	0.87	0.91	179
Walking	0.99	0.94	0.97	179
Accuracy		**0.81**		
Macro Avg	0.89	0.81	0.77	1260
Weighted Avg	0.89	0.81	0.77	1260

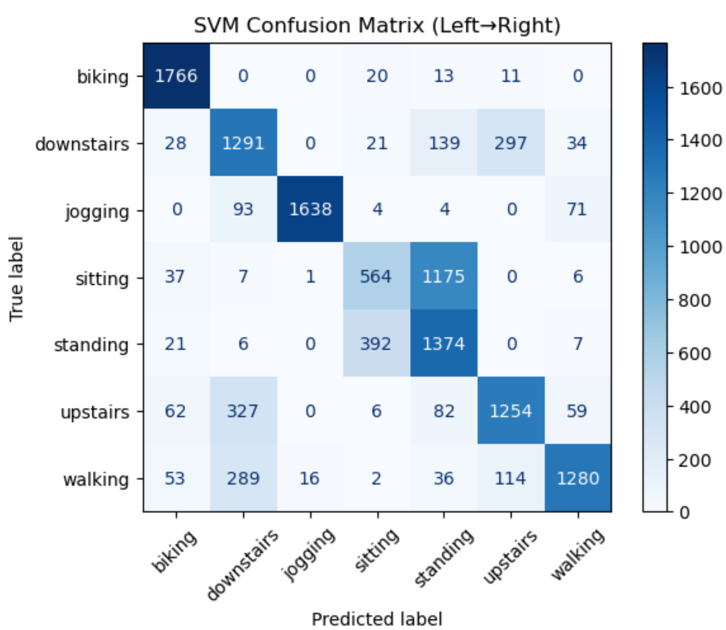


Fig. 15. Confusion matrix for the SVM classifier trained on left pocket data and tested on right pocket data, showing the true positive rates for each activity class across the LOSO folds. The diagonal values represent correct classifications, while off-diagonal values indicate misclassifications.

8 Discussion

Accuracy slightly would drop when testing on the wrist location whereas on pocket positions it performed in a similar way and missclassified the orientation of the static activities, highlighting the importance of sensor placement. Nevertheless, generalization performance remains satisfactory for pocket-based locations. The use of Random Forests enhances the robustness of the model, allowing it to handle variations in sensor orientation and placement. The interpretability of feature importance scores aids in understanding which aspects of the accelerometer signal contribute most to activity classification, guiding future feature engineering efforts. The insight of grouping similar activities and retraining the model to achieve higher accuracy comparing the two optimal classifiers (SVM and Random Forest) shows that the model can generalize better when activities are semantically similar, reducing the complexity of the classification task. The optimal model that we considered to be robust in most cases was the Random Forest classifier, especially considering the explainability, quick computational power and perfomance on seperate pocket position. Whereas the SVM classifier performed better on the grouped activities, it was more sensitive to hyperparameters and required more computational resources for training.

8.1 Key Results

Our pipeline demonstrates that robust HAR from a single smartphone accelerometer is feasible:

- **Random Forest** is the best performer (mean LOSO acc. = 0.78), with SVM and MLP not far behind.
- Adding spectral features (centroid, bandwidth, psd_sum) markedly improves class separation, especially between repetitive (jogging, biking) and static (sitting, standing) activities.
- Activity confusion is mainly seen among transitionally-similar classes (walking, stairs, standing/sitting).

All models generalized well under LOSO, suggesting patterns captured are not idiosyncratic.

9 Further Optimization and Recommendations

- **Advanced Features:** Consider wavelet transforms, autocorrelation, ARIMA models for fit and raw time-series modeling with deep learning (LSTM).
- **Ensemble Models:** Combining RF, SVM, and tuned MLP could yield a more robust “majority voting” classifier.
- **Multi-sensor Fusion:** Including gyroscope, magnetometer improves discriminatory power.
- **Consideration on Time Windows:** Experiment with different window sizes and overlaps to optimize feature extraction.

– **Confidence Intervals:** Use prediction intervals to quantify uncertainty in classifications.
– **Sensor Calibration:** If feasible, leverage calibration routines to adjust for individual posture.

10 Conclusions / Responding to the Research Questions

– We have shown that time and frequency features from ∼20s magnitude windows suffice to accurately classify common physical activities.
– Even classic ML models (RF, SVM, MLP) can achieve strong performance with good features and rigorous LOSO validation.
– The principal bottlenecks are between activities with overlapping dynamics, worsened by reduction to signal magnitude and lack of orientation information.
– Real-world deployment should restrict to learned locations or train on multi-location data; grouping similar classes can yield reliably high performance (85%+).

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